MD. ZOBAER AHMED // MZA CODE + LENS

Full-Stack Engineering × Documentary Photography Portfolio — Architecture Review

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Live URL: https://mdzobaerahmed.github.io/ | **Admin Portal:** #/admin
Review Date: October 2026 | **Version:** 2.0 (React 18, Vite 5, Tailwind CSS, Supabase)

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1. Executive Summary & Design Philosophy

The **MZA Code + Lens** portfolio represents a modern, high-performance web platform designed to harmonize two distinct disciplines: **Rigorous Computer Science / Product Engineering** and **Fine Art Documentary Photography (Fujifilm X-T5)**. The website delivers an immersive dark-aesthetic user experience (`#0B0F17` zinc-slate canvas, amber `#F59E0B` micro-accents, and monospace data typography), featuring instant client-side hash navigation, an interactive terminal simulator, an automated fine art photo gallery with EXIF camera telemetry, a password-protected administrator back-office protected by SHA-256 cryptographic hashing, and serverless direct-to-inbox messaging via Web3Forms.

2. Tech Stack & Libraries Breakdown

A comprehensive inventory of frameworks, toolchains, APIs, and utility libraries powering the production build:

Layer / Category	Technology / Library	Version	Architectural Purpose & Implementation Detail
Frontend Core	React 18	^18.3.1	Declarative UI architecture, functional component model, hooks (`useState`, `useEffect`), client-side hash router.
Build Engine	Vite 5	^5.4.21	Blazing fast Hot Module Replacement (HMR), Rollup-based tree-shaking production bundler generating optimized static assets.
Styling & Design	Tailwind CSS	^3.4.17	Utility-first styling with customized dark theme palette (`#0B0F17`), responsive breakpoints, custom scrollbars, and film grain textures.
Icon System	Lucide React	^1.16.0	Over 40 crisp vector icons (Camera, Terminal, Shield, Download, Lock, CheckCircle2, ShoppingBag, Eye, Mail, etc.).
Cloud & Database	Supabase Client	^2.49.1	PostgreSQL cloud storage & photo sync via `@supabase/supabase-js`, hybrid offline-first LocalStorage fallback.
Form Dispatch	Web3Forms API	REST API	Direct-to-inbox message forwarding to `mdzobaerahmed31@gmail.com` with zero exposed SMTP secrets; fallback to FormSubmit.
PDF Generation	jsPDF & html2canvas	^2.5.2 / ^1.4.1	Client-side dynamic generation of invoices and order receipts for photography print sales in the admin portal.
Security & Auth	Web Crypto API	Native SHA-256	Cryptographic SHA-256 hashing (`crypto.subtle.digest`) for administrator verification. Zero plain-text passwords in code.
Sanitization	DOMPurify	^3.2.4	XSS protection sanitizing rendered content, user messages, and dynamic markdown strings.
Hosting & CI/CD	GitHub Pages + Actions	v4 actions	Automated CI/CD pipeline (`.github/workflows/deploy.yml`) running Node 20 `npm run build` and zero-downtime publishing.

3. Page Architecture & Section-by-Section Breakdown

The portfolio operates on a responsive Single Page Architecture (SPA). A unified client-side hash listener (`window.location.hash`) manages both high-speed scrolling within the primary landing page and zero-latency sub-page rendering (`#/skills`, `#/academic-projects`, `#/admin`, etc.).

3.1 Navbar Component (`src/components/Navbar.jsx`)

- **Layout & Shell:** Fixed sticky header with blurred glass backdrop (`backdrop-blur-md bg-[#0B0F17]/85`), high z-index (50).
- **Branding:** Minimalist logo `■ MZA` with amber accents linking back to the home view.
- **Navigation Links & Menus:** Direct jump links to #about, #experience, #education, #projects, and #gallery.
- **Dropdown Menus:** Interactive dropdown for *Activities* (Skills, Academic Projects, Publications, Extra Curricular, Tech Blog) and *Photography* (Exhibition Gallery, Print Sale, Documentary, Admin Portal).

• **Controls:** Smooth Dark/Light mode theme toggle persisting state in `localStorage`, 'Let's Talk' contact CTA, and mobile hamburger drawer.

3.2 Hero Section (`src/components/Hero.jsx`)

• **Dual-Identity Header:** 'Full-Stack Engineering × Documentary Photography' positioning Md. Zobaer Ahmed as a CS undergrad and product manager who shoots fine-art photography.

• **Live Availability Badge:** Pulsing status indicator: 'Available for Summer 2026 PM / Tech Roles & Documentary Inquiries'.

• **Typography & Key Bio:** High-contrast heading (`text-4xl sm:text-6xl`), subtitle outlining United International University (UIU) 10th Trimester standing, and international photo curations in Greece & Scotland.

• **Dual Action CTAs:** 'Explore Tech & Engineering' (smooth scrolls to #activities) and 'View Fine Art Photography' (smooth scrolls to #gallery).

• **Interactive Terminal Card (`src/components/TerminalCard.jsx`):** Simulated developer terminal with clickable command buttons (`skills`, `experience`, `gear`, `contact`) and interactive CLI text output.

3.3 About & Career Journey Section (`src/components/AboutSection.jsx`)

• **Modular Section Tabs:** In-page filter pills toggling between All, Work Experience, Education, and Achievements.

• **01 // About Me:** Narrative biography detailing algorithmic problem solving, software engineering, digital product management, and visual storytelling.

• **02 // Professional Experience:** Structured timeline cards featuring IDP Education Ltd (IELTS Invigilator), UIU Photography Club (Vice President), Interactive Cares (Brand Representative), and Remains Art Club (Executive Member). Clean footer: Professional & Leadership Roles | 2020 — Present.

• **03 // Education & Academic History:** Comprehensive timeline: United International University (B.Sc. in CSE, 10th Trimester), Dhaka Residential Model College (HSC Science, GPA 5.00), and Chairman Bazar Secondary School (SSC Science). Clean footer: UIU CSE • DRMC.

• **04 // Credentials & Honors:** Certifications (Digital PM, Aspire Leaders Harvard Business School, Digital Marketing, Cybersecurity) and Awards (Life at UIU Winner, Reflection 3.0 Senior Winner, DRMC National Nature Festival, Government Merit Scholarships).

• **05 // Technical & Photographic Arsenal:** Side-by-side matrices comparing Software Engineering (C, C++, Python, JavaScript, PHP, MySQL, React, Node.js, Git) and Optics Gear (Fujifilm X-T5, Fujinon XF 35mm f/1.4 R, Adobe Lightroom, Classic Chrome simulation).

3.4 Academic & Engineering Projects (`src/components/EngineeringSection.jsx`)

• **Curated Showcase:** Featured high-impact software projects: **Workly** (Freelance Marketplace backend with Node.js, Express, TypeScript, Prisma), **OpportunityHub** (PHP/MySQL Scholarship and Job portal), **Multi-Client Chat System** (Python socket programming with multithreading), and **Livestock Manager** (System analysis & LaTeX).

• **Card Architecture:** Project category tags, system architecture highlights, technology badges, and live GitHub repository links.

• **Interactive Modal (`ProjectModal.jsx`):** Detailed system walkthrough, database schema description, and feature specifications upon clicking any project card.

3.5 Fine Art Photography Gallery (`src/components/PhotographyGallery.jsx`)

• **Exhibition-Grade Visual Grid:** Masonry / responsive grid showcasing authentic Fujifilm X-T5 photography in Dhaka and beyond.

• **Dynamic Filtering:** Real-time category selector pills: All, Street, Architecture, Nature, and Documentary.

• **EXIF Camera Telemetry:** Technical camera specs attached to every photo: Fujifilm X-T5, Fujinon XF 35mm f/1.4 R, 35mm, f/2.0, 1/250s, ISO 200, Classic Chrome.

• **Lightbox Modal (`LightboxModal.jsx`):** Fullscreen high-resolution modal with image zoom, contextual visual story, and quick print purchase action.

3.6 Contact & Communication Sub-Section

• **Web3Forms Direct Email Dispatch:** Contact form capturing Sender Name, Email, Subject, and Message. Submits directly via Web3Forms API to `mdzobaerahmed31@gmail.com` with real-time feedback states.

• **Direct Channels:** Clickable contact badges for Email, Phone (+8801783503006), Location (Dhaka, Bangladesh), LinkedIn, and GitHub.

• **Academic CV Download:** One-click button to download the formatted PDF CV (`Academic_CV.pdf`).

3.7 Footer Component (`src/components/Footer.jsx`)

• **Minimalist Technical Footer:** Copyright notice, tech stack acknowledgement ('Engineered with React 18, Vite & Tailwind CSS'), quick Back to Top smooth scroll, and external social links.

3.8 Dedicated Sub-Pages (Hash-Routed)

• **#/skills:** Deep dive technical competency matrix across programming languages, backend frameworks, databases, and DevOps.

• **#/academic-projects:** Extended catalog of academic coursework and laboratory implementations.

• **#/publications:** Research topics in machine learning, class imbalance, and computational health informatics.

• **#/extra-curricular:** Comprehensive leadership record managing 70+ members at UIU Photography Club and cultural festivals.

• **#/blog:** In-depth technical articles (e.g., 'Tackling Severe Class Imbalance with SMOTE in Clinical ML').

• **#/exhibitions:** Curated gallery features in European exhibitions across Greece and Scotland.

• **#/prints:** Fine art print collection with size selection (A4, A3, Exhibition 16x24), paper selection (Hahnemuhle Photo Rag), cart system, and USD/BDT currency toggle.

• **#/documentary:** Long-form visual documentary photo-essays.

• **#/admin:** Restricted administrator console for managing gallery photos, print orders, messages, and cloud settings.

4. Security Architecture & Administrator Access Control

To ensure complete data protection and maintain an uncompromised public repository on GitHub, the portfolio implements several key security mechanisms:

- **Cryptographic SHA-256 Hash Verification:** No plain-text administrator passwords exist anywhere in the source code or git history. The system hashes input strings using the browser's native crypto.subtle.digest('SHA-256') and compares them against a cryptographic hash (`6051fc84a...`).
- **Zero Plain-Text Hints:** The login interface (`#/admin`) displays zero default password hints or debugging clues. Incorrect authentication attempts return a generic 'Incorrect administrator password. Access denied.' message.
- **Session Isolation & LocalStorage Scope:** Administrator session authentication is preserved strictly in sessionStorage. Data modifications performed locally remain strictly within the client's browser sandbox and cannot modify the production codebase on GitHub.
- **Secretless Email Forwarding:** Contact form messages and print orders are dispatched via the Web3Forms public access token, routing emails directly to mdzobaerahmed31@gmail.com without exposing private mail server SMTP credentials or database connection strings.
- **Client-Side Hash Routing:** Hash-based routing (`#/`) guarantees 100% routing stability on static hosts like GitHub Pages without requiring server-side URL rewrite rules or causing 404 errors.

5. Main Component Code Listing (`src/App.jsx`)

The core orchestrator of the entire application (`src/App.jsx`), managing theme persistence, hash route changes, smooth anchor scrolling, and sub-page mounting:

```
import React, { useState, useEffect } from 'react';
import { Navbar } from './components/Navbar';
import { Hero } from './components/Hero';
import { EngineeringSection } from './components/EngineeringSection';
import { PhotographyGallery } from './components/PhotographyGallery';
import { AboutSection } from './components/AboutSection';
import { Footer } from './components/Footer';

// Dedicated Activity Pages
import { SkillsPage } from './pages/SkillsPage';
import { AcademicProjectsPage } from './pages/AcademicProjectsPage';
import { PublicationsPage } from './pages/PublicationsPage';
import { ExtraCurricularPage } from './pages/ExtraCurricularPage';
import { BlogPage } from './pages/BlogPage';

// Dedicated Photography Pages
import { ExhibitionGalleryPage } from './pages/ExhibitionGalleryPage';
import { PrintSalePage } from './pages/PrintSalePage';
import { DocumentaryPage } from './pages/DocumentaryPage';
import { AdminPage } from './pages/AdminPage';

export function App() {
  const [darkMode, setDarkMode] = useState(() => {
    const saved = localStorage.getItem('theme');
    if (saved) return saved === 'dark';
    return true; // default dark mode
  });

  // Filter states that can be toggled via Navbar dropdowns or in-page pills
  const [experienceTab, setExperienceTab] = useState('all'); // 'all', 'work', 'education'
  const [achievementsTab, setAchievementsTab] = useState('all'); // 'all', 'awards', 'certifications'
  const [activitiesTab, setActivitiesTab] = useState('all');

  // Page Routing State
  const [currentPage, setCurrentPage] = useState(() => {
    const hash = window.location.hash;
    if (hash.startsWith('#/')) {
      return hash.replace('#/', '');
    }
    return 'home';
  });

  useEffect(() => {
    const handleHashChange = () => {
      const hash = window.location.hash;
      if (hash.startsWith('#/')) {
        setCurrentPage(hash.replace('#/', ''));
        window.scrollTo({ top: 0, behavior: 'smooth' });
      } else {
        setCurrentPage('home');
```

```
      }
    };
    window.addEventListener('hashchange', handleHashChange);
    return () => window.removeEventListener('hashchange', handleHashChange);
  }, []);

  useEffect(() => {
    const root = document.documentElement;
    if (darkMode) {
      root.classList.add('dark');
      root.classList.remove('light');
      localStorage.setItem('theme', 'dark');
    } else {
      root.classList.remove('dark');
      root.classList.add('light');
      localStorage.setItem('theme', 'light');
    }
  }, [darkMode]);

  const navigate = (page) => {
    if (page === 'home') {
      window.location.hash = '';
      setCurrentPage('home');
      window.scrollTo({ top: 0, behavior: 'smooth' });
    } else {
      window.location.hash = `#/${page}`;
      setCurrentPage(page);
      window.scrollTo({ top: 0, behavior: 'smooth' });
    }
  };

  const navigateAndScroll = (targetId) => {
    if (currentPage !== 'home') {
      window.location.hash = '';
      setCurrentPage('home');
      setTimeout(() => {
        const el = document.querySelector(targetId);
        if (el) {
          el.scrollIntoView({ behavior: 'smooth' });
        }
      }, 100);
    } else {
      const el = document.querySelector(targetId);
      if (el) {
        el.scrollIntoView({ behavior: 'smooth' });
      }
    }
  };

  return (
    <div className={`min-h-screen transition-colors duration-300 film-grain ${
      darkMode ? 'bg-[#0B0F17] text-slate-100' : 'bg-[#FAFAFA] text-slate-900'
    }`}>
      {/* Top Navigation */}
      <Navbar
        darkMode={darkMode}
        setDarkMode={setDarkMode}
        currentPage={currentPage}
        navigate={navigate}
        navigateAndScroll={navigateAndScroll}
        experienceTab={experienceTab}
        setExperienceTab={setExperienceTab}
        achievementsTab={achievementsTab}
        setAchievementsTab={setAchievementsTab}
        activitiesTab={activitiesTab}
        setActivitiesTab={setActivitiesTab}
      />

      {/* Conditional Activity Page Rendering */}
      {currentPage === 'skills' && (
        <SkillsPage darkMode={darkMode} onBack={() => navigate('home')} />
      )}

      {currentPage === 'academic-projects' && (
        <AcademicProjectsPage darkMode={darkMode} onBack={() => navigate('home')} />
      )}

      {currentPage === 'publications' && (
        <PublicationsPage darkMode={darkMode} onBack={() => navigate('home')} />
      )}

      {currentPage === 'extra-curricular' && (
        <ExtraCurricularPage darkMode={darkMode} onBack={() => navigate('home')} />
```

```
      )}

      {currentPage === 'blog' && (
        <BlogPage darkMode={darkMode} onBack={() => navigate('home')} />
      )}

      {/* Conditional Photography Page Rendering */}
      {(currentPage === 'exhibition-gallery' || currentPage === 'exhibition' || currentPage === 'photography-gallery') && (
        <ExhibitionGalleryPage darkMode={darkMode} onBack={() => navigate('home')} />
      )}

      {currentPage === 'print-sale' && (
        <PrintSalePage darkMode={darkMode} onBack={() => navigate('home')} />
      )}

      {currentPage === 'documentary' && (
        <DocumentaryPage darkMode={darkMode} onBack={() => navigate('home')} />
      )}

      {/* Admin Management Dashboard */}
      {currentPage === 'admin' && (
        <AdminPage darkMode={darkMode} onBack={() => navigate('home')} />
      )}

      {/* Main Home Portfolio View */}
      {currentPage === 'home' && (
        <>
          {/* Hero Section */}
          <Hero
            darkMode={darkMode}
            onExploreCode={() => navigateAndScroll('#activities')}
            onExploreLens={() => navigateAndScroll('#gallery')}
          />

          {/* Main Content Sections In User's Exact Order */}
          <main className="space-y-16">
            {/* 01-04: About Me, Work Experience, Education, Achievements */}
            <AboutSection
              darkMode={darkMode}
              experienceTab={experienceTab}
              setExperienceTab={setExperienceTab}
              achievementsTab={achievementsTab}
              setAchievementsTab={setAchievementsTab}
              activitiesTab={activitiesTab}
              setActivitiesTab={setActivitiesTab}
              onNavigate={navigate}
              navigateAndScroll={navigateAndScroll}
            />

            {/* 05: Activities // Academic Projects */}
            <EngineeringSection
              darkMode={darkMode}
              onNavigateToAcademicProjects={() => navigate('academic-projects')}
              onNavigate={navigate}
            />

            {/* 06: Photo Gallery // Exhibition, Print Sale, Documentary */}
            <PhotographyGallery
              darkMode={darkMode}
              onNavigateToExhibition={() => navigate('exhibition-gallery')}
              onNavigate={navigate}
            />
          </main>
        </>
      )}

      {/* Footer & Contact */}
      <Footer darkMode={darkMode} onNavigate={navigate} />
    </div>
  );
}

export default App;
```